Homework 11

Massively Parallel 5’ Splice Site Analysis

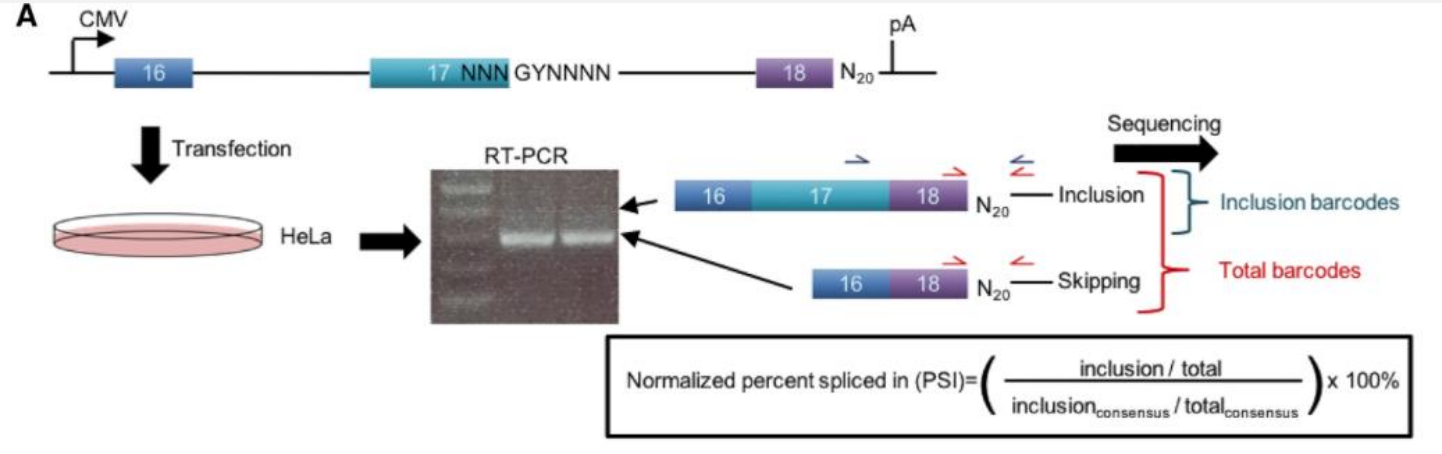
Background

- MPSA to look into splicing at BRCA2's intron 17 locus.
  - Focus on the non-canonical BRCA2 splicing locus (GC instead of GU)
- Sequencing readout-
  - Inclusion barcodes (include exon 17 aka proper splicing)
  - Total barcodes
- Using equation, they calculate PSI (metric used for this homework's analysis)

> Mol Cell. 2018 Sep 20;71(6):1012-1026.e3. doi: 10.1016/j.molcel.2018.07.033. Epub 2018 Aug 30.

Quantitative Activity Profile and Context Dependence of All Human 5' Splice Sites

Mandy S Wong [1], Justin B Kinney [2], Adrian R Krainer [3]



Wong et al., 2018

Instructions

- Use the files below –
  - mpsa_analysis.py (python script to complete)
  - mpsa_data.csv (MPSA data file)
  - brca2_mutations.txt (Text file of sequences of pathogenic BRCA2 splice site mutations)
- Complete the code in the python script to answer the questions
  - You may deviate from the comments and suggestions in the code if you want to do it a different way
- Suggestion- Use something like Jupyter notebooks or VS code to test your code and run it
- Note- script should be able to run independently on the server

Task 1 – Complete helper functions

- MPSA sequences are 9nt long and begin at -3 position.
- In order to use them for downstream analysis, we want to convert them to more typical string positions from 0-8
- *ss_to_string()* function
  - Converts 5' Splice Site positions to python string positions
  - It should take the splice site position as an input, and return the corresponding 'python string position' as given by
  - Professor White provided a few lines of code for you to test your function!

```python
def ss_to_string(ss_position):
    '''Convert splice site position (-3 to +6, no zero)
    to strong position (0 to 8)'''
    # The -3 site is at string position 0.
    # Reject positions that are before -3:
    if ss_position < -3:
        print('Out of range')
        return None
    # Complete the function below:
```

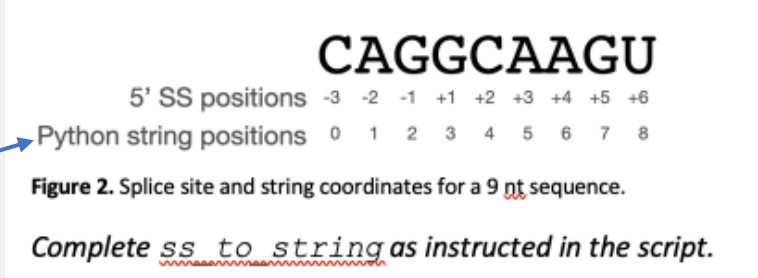


Figure 2. Splice site and string coordinates for a 9 nt sequence.

Complete `ss_to_string` as instructed in the script.

Task 2 – Examine distribution of PSI values

- We want to examine the distribution of PSI values across the dataset.
- Do this by splitting the entire list of PSI values into ranges of 0-10%, 10-20% and so on.
  - You are provided the function *get_decile_index()* that you can use to determine the decile index of each PSI value.
  - You then need to count the number of times we observe a PSI value for each index.
- You do not need to make a histogram (but are welcome to), looking at the 10 indices should be sufficient to answer the question

```python
# A function to determine PSI decile:
def get_decile_index(psi, minumum = 0, bin_width = 10):
    '''Returns the decile  for measurements scaled 0 to 100. PSI of
    >= 100 is assigned the last decile.'''
    index = int((psi - minumum) // bin_width)
    if index > 9:
        index = 9
    return index
```

Task 3 + 4 – Examine distribution of ‘GC’ splice sites

- Earlier we looked at the PSI distribution of all splice sites (GC and GU)
- Now focus on the non-canonical GC splice sites.
- Comment on the PSI distribution of GC splice sites

- Examine the sequences in the top decile focusing on positions of interest determined from the paper (-1 and +5)

Task 5 – Examine pathogenic BRCA2 splice variants

- Determine the splicing efficiency of known BRCA2 pathogenic variants
- Read in the list of known pathogenic BRCA2 splice variants (brca_mutations.txt)
- Find the splice variants in the dataframe and pull their corresponding PSI values
- Determine the distribution and mean of PSI values.

What to turn in:

- Completed mpsa_analysis.py script
- A README.txt file with answers to the questions
- The output.txt file obtained by running the mpsa_analysis.py script